p-adic Analysis 8

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Problem 1

No. Assume $x \in \widehat{\mathbb{Z}}$. Then for all n, $2x_{2n} \equiv 0 \pmod{2n}$. So x_{2n} is either $0 \pmod{n}$ or $n \pmod{2n}$. That means $x_n \equiv 0 \pmod{n}$ in both cases, so $x = 0$. Therefore 2 is not a zero divisor in $\widehat{\mathbb{Z}}$.

Problem 5

We just need $x_p = x_p^2$ for all primes p, or $x_p(x_p - 1) = 0$. Since $x_p \in \mathbb{Z}_p$, this forces $x_p = 0$ or 1 for all p. Therefore all $x = (x_2, x_3, x_5, \ldots) \in \widehat{\mathbb{Z}}$ with $x_i = 0$ or 1 satisfy $x^2 = x$. The size of this set is $2^{\aleph_0}$.

Problem 6

Since

$$\mathbb{Z}_m = \prod_{p|m} \mathbb{Z}_p,$$

the number of solutions to $x^3 = x$ is the product of the number of solutions in each $\mathbb{Z}_p$. Also, $x^3 = x$ iff $x(x-1)(x+1) = 0$. So if p is odd, there are 3 solutions $(0, 1, -1)$, and if $p = 2$ there are two solutions $(0, 1)$. So each odd prime factor of m gives 3 choices and a factor of 2 gives 2 choices.

Problem 8

Let $s_N = \sum_{n=0}^{N} a_n$, and assume s_N is Cauchy. That means we have $|s_N - s_{N-1}|_m \to 0$. But since $a_N = s_N - s_{N-1}$, we get $|a_N|_m \to 0$.

Now, let $|a_n|_m \to 0$. For all $N > M$, we have that

$$s_N - s_M = a_{M+1} + a_{M+2} + \cdots + a_N.$$

By the ultrametric triangle inequality,

$$|s_N - s_M|_m \leq \max_{M<k\leq N} |a_k|_m.$$

Since $|a_k|_m \to 0$, $|s_N - s_M|_m \to 0$, and therefore s_N is Cauchy.

Problem 10

From problem 1, 2 is not a zero divisor. Also, it is not a unit because for there to be some y where $2y = 1$, reducing mod 2 gives us $0 \equiv 1 \pmod 2$, which is impossible.